

A Project Report on
Sign language recognition using Sign-IDD model

Summer Internship Report

submitted in partial fulfilment of the requirements for the award

of the degree of

BACHELOR OF TECHNOLOGY
in
Computer Science and Engineering

Submitted by
Akash Yadav
(Roll no: CS-23411217)

Under the supervision of
Ms. Surabhi Purwar
Assistant Professor



Intern Details:

- **Intern Name:** Akash Yadav
- **Roll No.:** CS-23411217
- **Internship Organization:** CSIR - 4PI
- **Duration:** April 2026 – July 2026

IILM University, Greater Noida, Uttar Pradesh

Academic Session (2026-27)

CANDIDATE'S DECLARATION

I, Akash Yadav do hereby declare that the internship report titled “ ***Sign language recognition using sign-IDD model*** ” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on my internship learning and project work. I have acknowledged the sources and learning materials used in preparing this report. I affirm that this report has not been submitted to any other institute for the fulfilment of any degree/diploma requirement and I shall be responsible for any copyright violation arising from material submitted by me.

Signature: Akash Yadav

Student Name: Akash Yadav

Roll No.: CS-23411217

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **CSIR-Fourth Paradigm Institute (CSIR-4PI)** for providing me the opportunity to undertake this internship and work on the project **Sign Language Recognition using the Sign-IDD Model**. This experience has been highly enriching and has significantly strengthened my technical and research skills.

I also express my gratitude to my faculty members and mentors at IILM University for their guidance, encouragement and support throughout the internship work. I am thankful to my parents and everyone who supported me during the completion of this internship and preparation of this report.

Signature: Akash Yadav

Student Name: Akash Yadav

Roll No.: CS-23411217

CONTENTS

1. Introduction
 2. Literature Review
 3. Objectives
 4. Tools And Technologies Used
 5. Methodology
 6. Implementation
 7. Results And Discussion
 8. Conclusion
 9. Future Scope
 10. Internship Learning Outcomes
- References

Introduction

Communication is a fundamental part of human life, allowing people to share information, express feelings, and interact meaningfully. For individuals who are hearing or speech impaired, verbal communication is not a natural option, and they rely on sign language as their primary method of expression. Sign language uses a structured combination of hand gestures, facial expressions, and body movements to represent words and sentences. Although this form of communication is highly expressive and efficient within the deaf community, the majority of the general population does not understand it. This lack of common understanding creates a communication barrier, making it difficult for hearing-impaired individuals to interact smoothly in social spaces, workplaces, classrooms, and public environments.

To address this challenge, sign language recognition has become an important research area in artificial intelligence and computer vision. These systems are designed to automatically analyze sign gestures and convert them into readable text or speech, enabling seamless communication between signers and non-signers. The central problem addressed in this project is the development of a model capable of accurately recognizing **Indian Sign Language (ISL)** gestures from video recordings and translating them into meaningful textual output. The aim is to create an accessible system that reduces communication gaps and provides support to hearing-impaired individuals in everyday scenarios.

The objective of this internship project is to build a machine-learning-based sign language recognition system by combining **pose estimation** and **deep learning**. Instead of relying on raw image pixels, the project focuses on extracting **2D skeletal keypoints** from sign language videos using **OpenPose**, an open-source real-time multi-person pose estimation framework. By using skeletal keypoints, the system becomes more robust to variations in background, lighting, and signer appearance. The project uses the **Sign-IDD dataset**, a publicly available dataset for Indian Sign Language recognition that contains labeled videos of various gesture classes.

The workflow begins with studying and organizing the Sign-IDD dataset. The raw video files are preprocessed using tools such as **FFmpeg**, ensuring consistent frame rate and format. Each processed video is then passed through **OpenPose**, which extracts 2D coordinates of important body joints and hand landmarks. These keypoints capture essential information about hand orientation, finger movement, and upper body posture—features necessary for distinguishing between different gestures.

LITERATURE REVIEW

To evaluate the system through repeated testing and observation of recognition performance.

- To develop an LSTM-based deep learning model for gesture classification.
- To normalize and transform skeletal data into suitable numerical sequences.
- To generate structured .skel files containing frame-wise keypoint information.
- To extract 2D body and hand skeletal keypoints using OpenPose.
- To preprocess videos using FFmpeg for consistent frame rate and format.
- To study and organize the Sign-IDD dataset containing labeled ISL gesture videos.
- To develop a machine-learning-based system for recognizing Indian Sign Language gestures from video sequences.

Objective

.skel files – used to store frame-wise 2D skeleton coordinates in structured form.

- LSTM – used to learn temporal dependencies in the sequence of skeletal keypoints.
- OpenCV – used for image/video processing and supporting computer-vision operations.
- NumPy – used for numerical processing and preparation of skeleton sequences.
- PyTorch – used for developing and training the LSTM-based deep learning model.
- FFmpeg – used for video preprocessing and conversion into a consistent format.
- Sign-IDD Dataset – used as the source of labeled Indian Sign Language gesture videos.
- OpenPose – used for extracting 2D body and hand keypoints from video frames.
- Python – used for data processing, feature preparation and model development.

Tools and Technologies Used

The project was developed using various tools and technologies required for video preprocessing, skeleton extraction, feature preparation and deep learning-based gesture recognition.

- **Python:** Used for data processing, feature preparation and implementation of the machine learning pipeline.
- **OpenPose:** Used to extract 2D body and hand keypoints from the sign language videos.
- **Sign-IDD Dataset:** Used as the primary dataset containing labeled Indian Sign Language gesture videos.
- **FFmpeg:** Used for converting and preprocessing videos into a consistent format.
- **PyTorch:** Used for developing and training the LSTM-based deep learning model.
- **NumPy:** Used for numerical operations and processing skeletal coordinate data.
- **OpenCV:** Used for video and image processing operations.
- **LSTM (Long Short-Term Memory):** Used to learn temporal patterns from the sequence of skeletal keypoints.
- **.skel Files:** Used to store frame-wise 2D skeletal keypoint information extracted from the videos.

Methodology

The development process follows a sequential pipeline consisting of dataset preparation, video preprocessing, skeleton extraction, feature processing, model development and testing.

Step 1 – Dataset Preparation:

The Sign-IDD dataset was studied and organized according to the available Indian Sign Language gesture classes.

Step 2 – Video Preprocessing:

The input videos were converted into a consistent format using FFmpeg, maintaining suitable frame rate and resolution for further processing.

Step 3 – Skeleton Extraction:

OpenPose was used to extract 2D body and hand keypoints from each video frame. The extracted keypoints were stored in .skel files.

Step 4 – Feature Processing:

The extracted skeleton coordinates were normalized to reduce variations caused by signer height, body proportions and camera position. Temporal motion features were also obtained from frame-to-frame changes in joint positions.

Step 5 – Model Development:

The processed skeleton sequences were used to develop an LSTM-based deep learning model using PyTorch. The LSTM model was selected to learn the temporal dependencies present in gesture sequences.

Step 6 – Model Training and Evaluation:

The dataset was divided into training, validation and testing subsets. The model was trained using the prepared skeleton sequences and evaluated based on its gesture recognition performance.

Step 7 – Testing:

New video inputs were processed through OpenPose to generate .skel files, which were then provided to the trained LSTM model to obtain the predicted gesture label.

Results and Discussion

The developed system successfully processed the Sign-IDD dataset through video preprocessing, OpenPose-based skeleton extraction, feature normalization, and LSTM-based gesture classification. OpenPose generated 2D body and hand keypoints, which were stored in .skel files and used for model training and testing.

The LSTM model produced stable and consistent predictions for most of the selected Indian Sign Language gestures. Gestures with clear hand movements and distinct body trajectories were recognized more reliably. The average processing time was approximately **1.5–2 minutes per video** on CPU.

The results also showed that video quality affected recognition performance. Good lighting and clear hand visibility produced better keypoints, while motion blur and hand occlusion sometimes caused incorrect predictions.

Some misclassifications occurred between visually similar gestures due to small differences in finger and wrist positions. The use of 2D keypoints also limited the system in capturing depth-based movements. Overall, the results demonstrate that combining **OpenPose 2D keypoints with LSTM** is a practical approach for ISL gesture recognition.

Conclusion

The project successfully developed a sign language recognition system using the Sign-IDD dataset, OpenPose, 2D skeletal keypoints, and an LSTM-based deep learning model. The system was able to recognize selected Indian Sign Language gestures and produce stable predictions during testing.

Although GPU compatibility issues required the system to operate on CPU, the overall pipeline performed reliably. The results show that combining skeletal keypoint extraction with temporal deep

learning is an effective approach for ISL gesture recognition. Future improvements can focus on GPU acceleration, 3D keypoints, larger datasets, and real-time gesture recognition.

Future Scope

The developed system can be further improved to make it faster, more accurate, and suitable for real-world applications. Future work can include:

- **GPU acceleration** to reduce skeleton extraction and processing time.
- **3D pose estimation** to capture depth-based hand and body movements.
- **Larger and more diverse datasets** to improve recognition across different users and environments.
- **Real-time webcam integration** for live sign language recognition.
- **Continuous sentence-level recognition** to recognize complete sequences of signs instead of individual gestures.
- **Mobile or assistive-device deployment** to make the system more accessible for hearing- and speech-impaired individuals.

INTERNSHIP LEARNING OUTCOMES

During the internship, I gained practical knowledge and hands-on experience in the following areas:

- Learned how to work with the Sign-IDD dataset and organize labeled sign language videos.
- Gained practical experience in video preprocessing using FFmpeg.
- Learned to extract 2D body and hand keypoints using OpenPose.
- Understood the process of generating and handling .skel files containing skeletal data.
- Gained experience in data normalization and temporal feature processing.
- Learned to develop and train an LSTM-based deep learning model using PyTorch.
- Developed an understanding of model testing, gesture prediction, and performance evaluation.
- Learned to identify and handle practical challenges such as noisy keypoints, occlusion, and GPU compatibility issues.
- Improved practical skills in Python, OpenCV, NumPy, and deep learning-based computer vision.

References

1. OpenPose – CMU Perceptual Computing Lab
<https://github.com/CMU-Perceptual-Computing-Lab/openpose>
2. Sign-IDD Dataset
<https://github.com/NaVi-start/Sign-IDD>
3. OpenCV Documentation
<https://opencv.org/>

4. RWTH-PHOENIX-Weather 2014: Continuous Sign Language Recognition Dataset

<https://www-i6.informatik.rwth-aachen.de/~koller/RWTH-PHOENIX/>

5. Research Papers:

- a. Progressive Transformers for End-to-End Sign Language Production

<https://arxiv.org/pdf/2004.14874>

- b. OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields

<https://arxiv.org/abs/1611.08050>

6. OpenPose Sign Language Recognition – Google Colab Notebook

https://colab.research.google.com/github/changsin/DL/blob/main/notebooks/openpose_sign_language.ipynb

Implementation & Results

The proposed Sign Language Recognition system was implemented using Python, OpenPose, PyTorch, NumPy, OpenCV, and FFmpeg. The Sign-IDD dataset was first preprocessed using FFmpeg, after which OpenPose was used to extract 2D body and hand keypoints. The extracted keypoints were stored in .skel files and normalized before being used as input to the LSTM model.

The LSTM model was trained using the processed skeletal sequences to learn the temporal patterns of different ISL gestures. During testing, new videos were processed through OpenPose and the generated skeleton sequences were given to the trained model for gesture prediction.

The system produced stable and consistent predictions for most selected gestures. The average processing time was approximately **1.5–2 minutes per video** on CPU. Better results were observed for videos with clear hand visibility and stable lighting, while motion blur and occlusion sometimes caused incorrect predictions.

Overall, the implementation successfully demonstrated that **2D skeletal keypoints combined with LSTM-based temporal learning can be effectively used for Indian Sign Language gesture recognition.**

